Is the Dataset Biased?

Leave space for headshot

Responsible AI: Human-Centered Design

START → DESIGN → **DEVELOPMENT** → DEPLOYMENT → END

Course 1
Fundamentals of TinyML

- **What** am I building?
- **Who** am I building this for?
- What are the **consequences** for the user if it ***fails***?

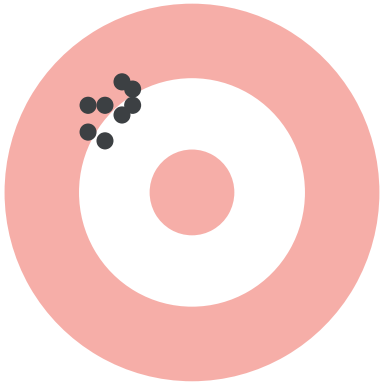
Course 2
Applications of TinyML

- What data will be collected to train the model?
- **Is the dataset biased?**
- How can we ensure the model is fair?

Course 3
Deploying TinyML

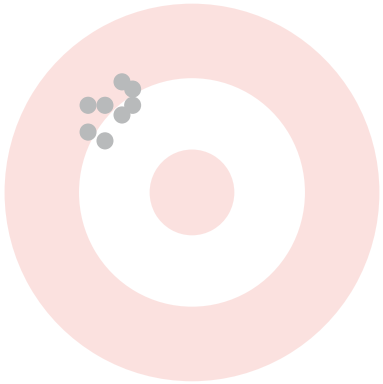
What is bias?

Bias is a deviation in a ***predictable*** (i.e., not random) direction

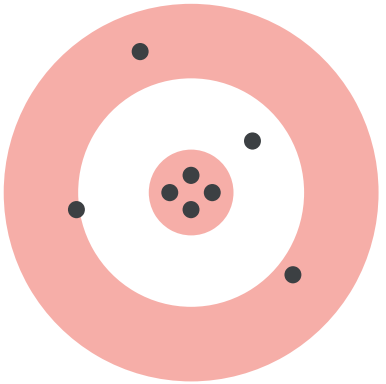


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Not all errors are attributed to bias



What does **bias** mean for **ML**?

Bias in machine learning = **systematic** errors that lead to inaccurate results

- Biased **datasets** (faulty, poor, or incomplete data) can lead to a distribution **mismatch** between the ***dataset*** and ***reality***
- This may lead to **inaccurate results**, or worse, **discriminatory** or **unfair results**

The **“garbage in, garbage out”** problem



Bias: Defining the **Target Variable**

- **How** should you define a ***“good”*** employee for a hiring algorithm?
- **Subjective** process: ***“good”*** must be defined in ways that correspond to **measurable** outcomes

Number of sick days	**Number of times late to work**
Length of employment	**Number of sales per quarter**

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Bias: Defining the **Target Variable**

Using **biometric** sensors for a health wearable device, how should you define ***“healthy”***?

- **Heart rate**
- **Blood pressure**
- **Number of steps**



Bias: **Labeling** the Data

Labels applied to the training data must serve as **ground truth**



Horse



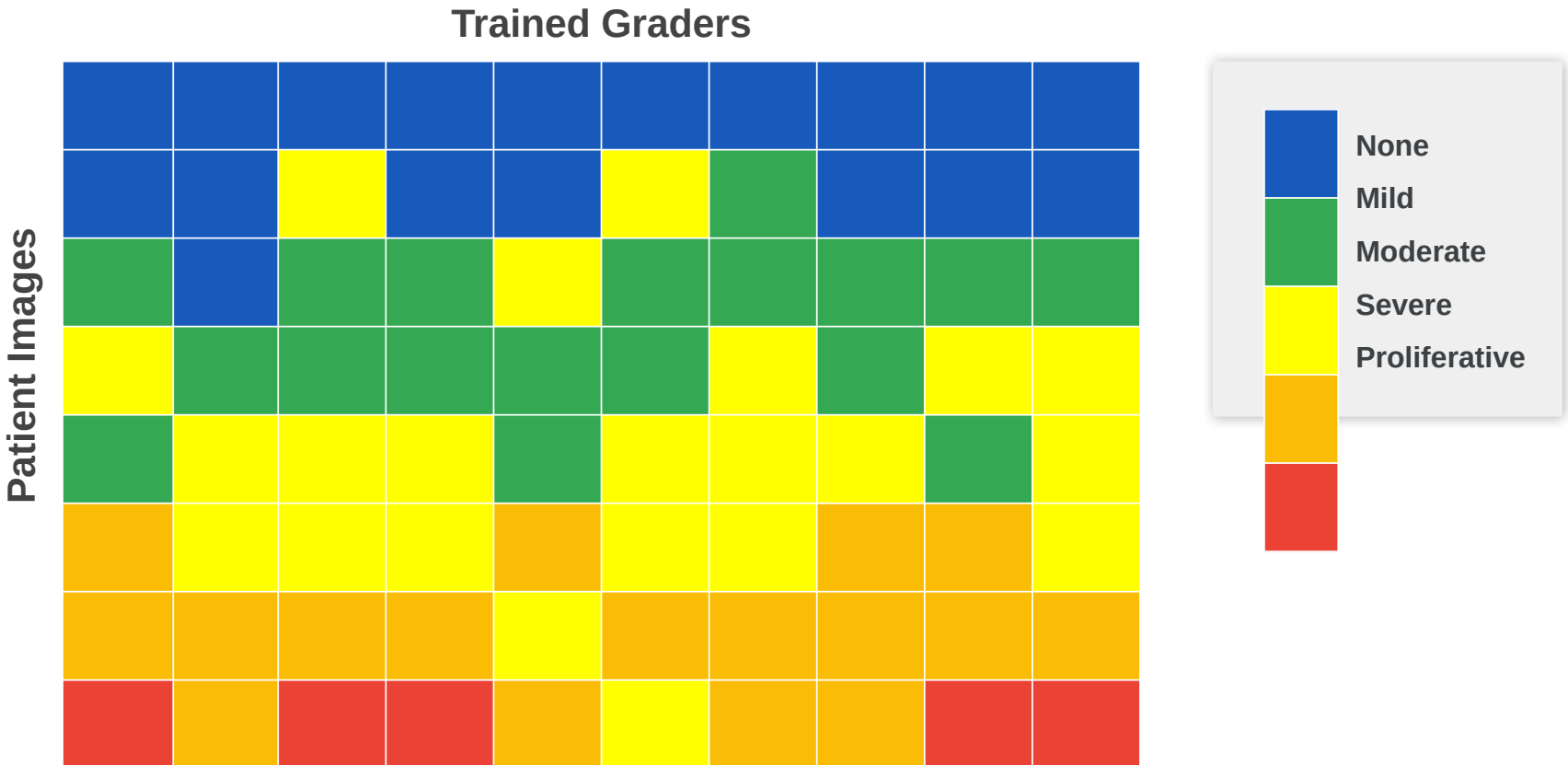
Human



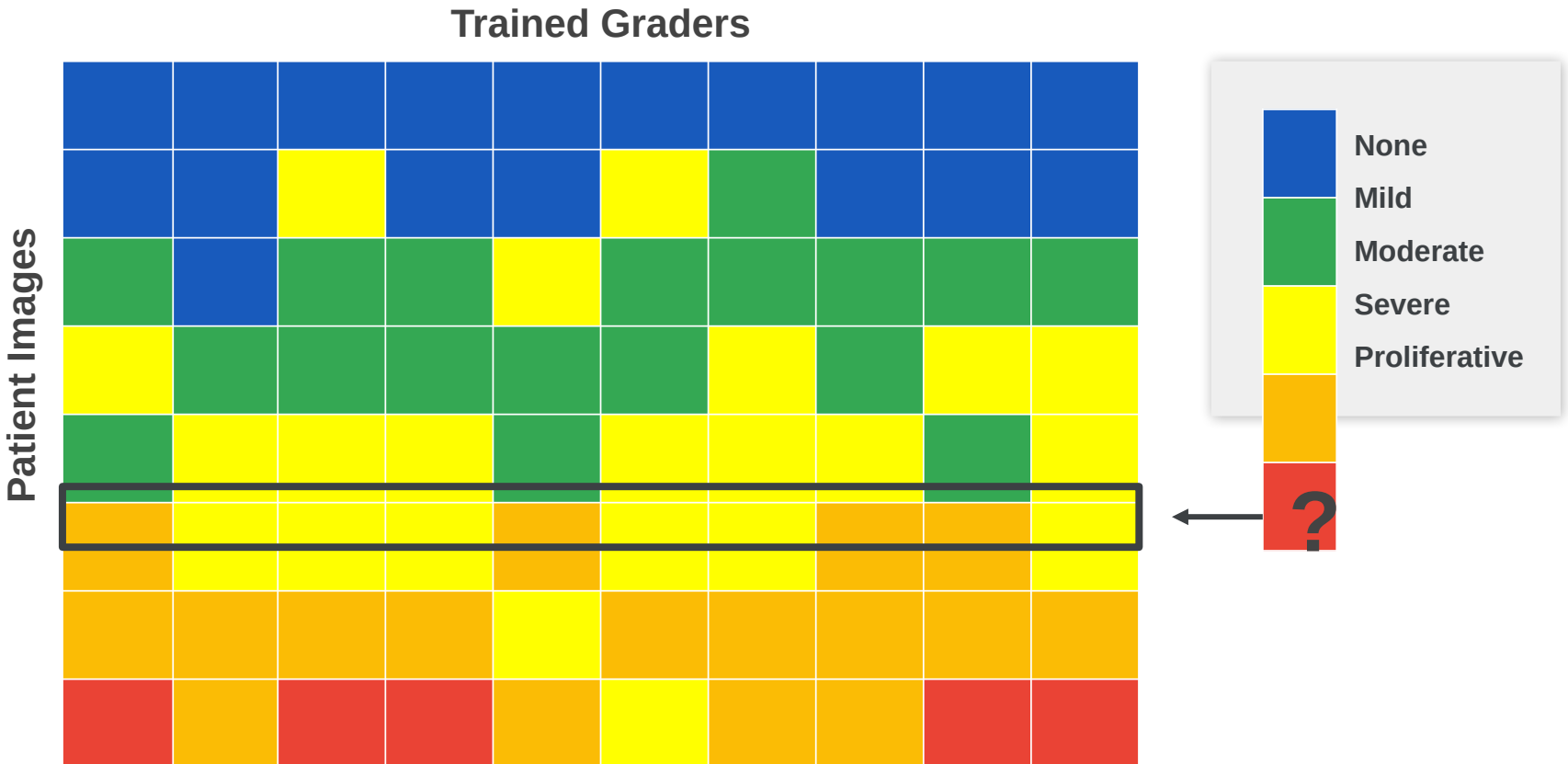
Human

ERRO
R

Bias: **Labeling** the Data



Bias: **Labeling** the Data



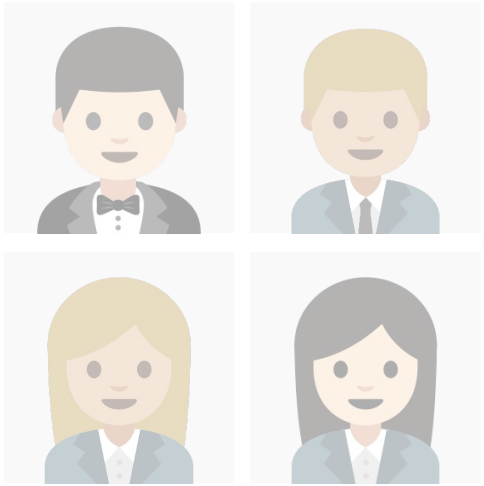
Bias: **Sampling** the Data

Training Data



Bias: **Sampling** the Data

Training Data



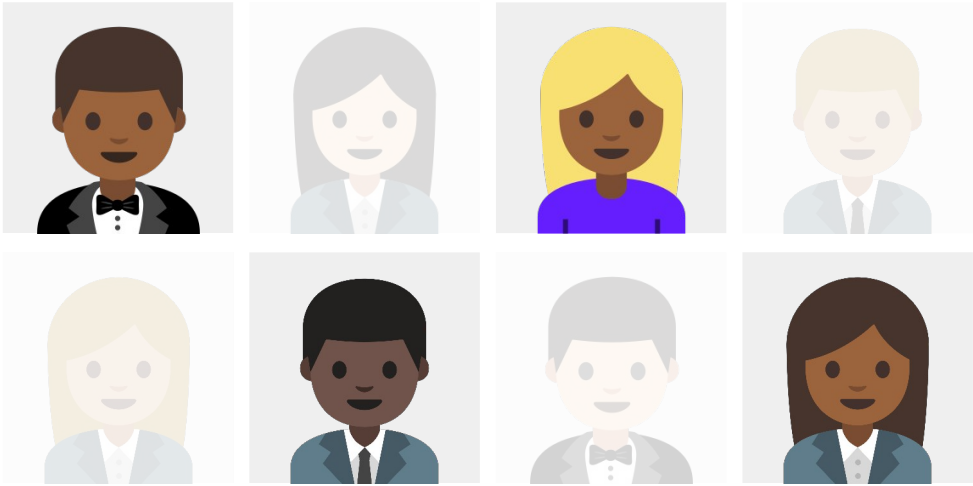
Reality



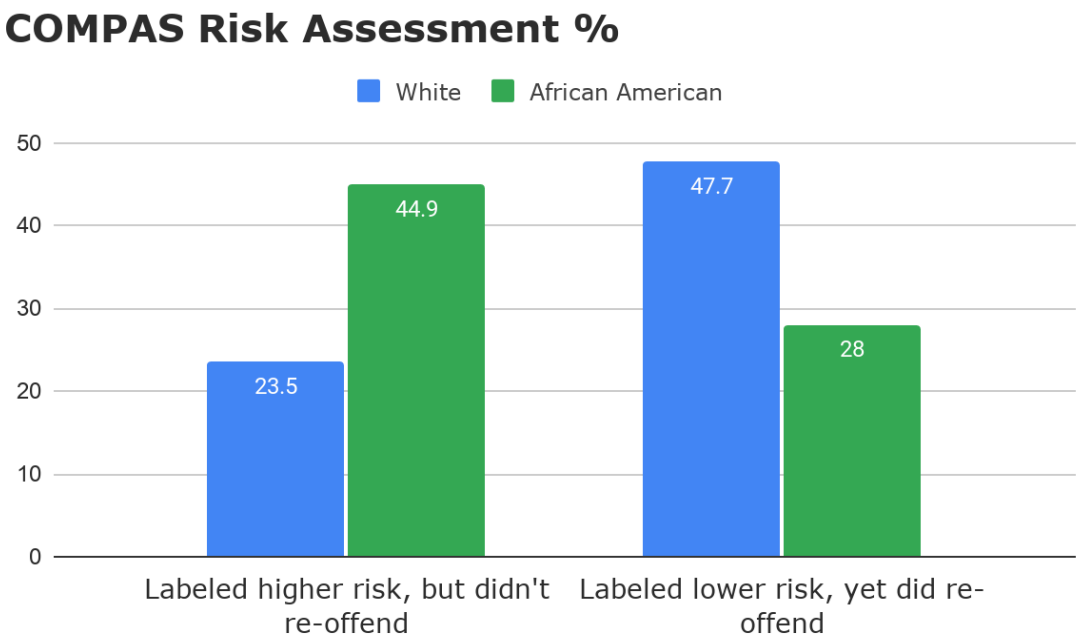
Bias: **Sampling** the Data

Training Data

Reality



Bias: **Prejudice** Reflected in Data



Northpointe's COMPAS Recidivism Prediction Tool

Bias: **Prejudice** Reflected in Data





Dataset: *65%* of people cooking are *women*

Algorithm predicts: *85%* of people cooking are *women*

Bias: Measurement **Distortion**



Hardware matters!

Example: Optic sensors and cameras have problems ***detecting*** ***darker skin tones*** due to the way light is reflected

- **Automatic soap dispensers**
- **Activity detection**

Bias: Measurement **Distortion**



How can we ***fix*** biased data?

Industry Solutions: **Datasheets for Datasets**

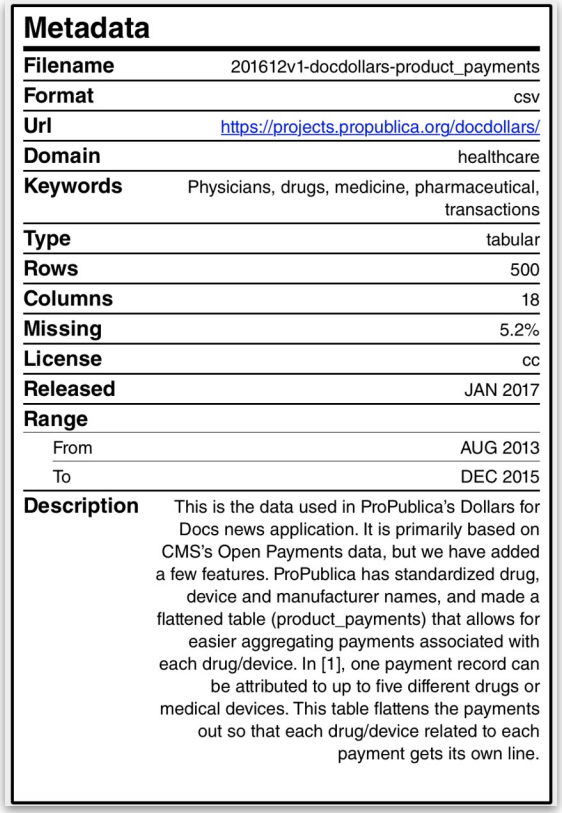
Questions for dataset creators to reflect on during the key stages of the dataset lifecycle:

- ***Motivation***
- ***Composition***
- ***Collection Process***
- ***Preprocessing/ labeling***
- ***Uses***
- ***Distribution***
- ***Maintenance***

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Industry Solutions: **Data Nutrition Labels**



Metadata	
Filename	201612v1-docdollars-product_payments
Format	csv
Url	https://projects.propublica.org/docdollars/
Domain	healthcare
Keywords	Physicians, drugs, medicine, pharmaceutical, transactions
Type	tabular
Rows	500
Columns	18
Missing	5.2%
License	cc
Released	JAN 2017
Range	
From	AUG 2013
To	DEC 2015
Description	This is the data used in ProPublica's Dollars for Docs news application. It is primarily based on CMS's Open Payments data, but we have added a few features. ProPublica has standardized drug, device and manufacturer names, and made a flattened table (product_payments) that allows for easier aggregating payments associated with each drug/device. In [1], one payment record can be attributed to up to five different drugs or medical devices. This table flattens the payments out so that each drug/device related to each payment gets its own line.

DATA NUTRITION PROJECT

A standard label that highlights the **"key ingredients"** of a dataset:

- ***Provenance***
- ***Metadata***
- ***Missing units***
- ***Variables***

Industry Solutions: **Bias Testing Toolkits**

IBM Research Trusted AI

AI Fairness 360

This extensible open source toolkit can help you examine, report, and mitigate discrimination and bias in machine learning models throughout the AI application lifecycle. We invite you to use and improve it.



Designer Solutions

- Carefully **research your users in advance**, be aware of potential outliers
- **Ensure your team** of data scientists and data labelers is ***diverse***
- Where possible, **combine inputs from multiple sources** to ensure data diversity
- **Create a gold standard** for data labeling
- Seek out **domain experts** to review your data